

Sage Research Methods

Doing Triangulation and Mixed Methods

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Front Matter

- Copyright
- List of illustrations
- Figures
- Tables
- Editorial introduction
- About this book

Chapters

- **Chapter 1 |** Why triangulation and mixed methods in qualitative research?
- **Chapter 2 |** What is triangulation?
- **Chapter 3 |** Methodological triangulation in qualitative research
- **Chapter 4 |** Triangulation in ethnography
- **Chapter 5 |** What is mixed methods research?
- **Chapter 6 |** Designs, methods and data in mixed methods research
- **Chapter 7 |** Triangulation as a framework for using mixed methods
- **Chapter 8 |** How to use triangulation and mixed methods in qualitative research: practical issues
- **Chapter 9 |** Sensitive use of multiple methods: quality, writing, and ethics

Back Matter

- Glossary
- References

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List of illustrations

Boxes

2.1 Definition of triangulation 23
3.1 Example of an interview guide for an episodic interview 31
3.2 Excerpts from the interview guide for reconstructing a subjective theory 41

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Figures

3.1 Within-method triangulation 26
3.2 Areas of everyday knowledge in the episodic interview 29
3.3 Data sorts in the episodic interview 35
3.4 Triangulation of different qualitative methods 38
3.5 Excerpt from a subjective theory of trust 42
6.1 Research designs for the integration of qualitative and quantitative research 81
8.1 Levels of triangulation of qualitative and quantitative research 122

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Tables

2.1 Comprehensive triangulation 19
2.2 Systematic triangulation of perspectives 22
4.1 Utilization patterns of medical services 66
4.2 Sample for the expert interviews 67
4.3 Health care deficits seen by the experts 68
8.1 Dimensions of comparison in studies using triangulation 115

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Editorial introduction

Introduction to *The SAGE Qualitative Research Kit*

In recent years, qualitative research has enjoyed a period of unprecedented growth and diversification as it has become an established and respected research approach across a variety of disciplines and contexts. An increasing number of students, teachers and practitioners are facing questions and problems of how to do qualitative research – in general and for their specific individual purposes. To answer these questions, and to address such practical problems on a how-to-do level, is the main purpose of *The SAGE Qualitative Research Kit*.

The books in *The SAGE Qualitative Research Kit* collectively address the core issues that arise when we actually do qualitative research. Each book focuses on key methods (e.g. interviews or focus groups) or materials (e.g. visual data or discourse) that are used for studying the social world in qualitative terms. Moreover, the books in the *Kit* have been written with the needs of many different types of reader in mind. As such, the *Kit* and the individual books will be of use to a wide variety of users:

- *Practitioners* of qualitative research in the social sciences; medical research; marketing research; evaluation; organizational, business and management studies; cognitive science; etc., who face the problem of planning and conducting a specific study using qualitative methods.
- *University teachers* and lecturers in these fields using qualitative methods can use this series as a basis of their teaching.
- *Undergraduate and graduate students* of social sciences, nursing, education, psychology and other fields where qualitative methods are a (main) part of the university training including practical applications (e.g. when writing a thesis).

Each book in *The SAGE Qualitative Research Kit* has been written by distinguished authors with extensive experience in their field and in practice with the methods they write about. When reading the whole series of books from the beginning to the end, you will repeatedly come across some issues which are central to any sort of qualitative research – such as ethics, designing research or assessing quality. However, in each book such issues are addressed from the specific methodological angle of the authors and the approach they describe. Thus you may find different approaches to issues of quality or different suggestions for how to analyze qualitative data in different books, which will combine to present a comprehensive picture of the field

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as a whole.

What is qualitative research?

It has become more and more difficult to find a common definition of qualitative research which is accepted by the majority of qualitative research approaches and researchers. Qualitative research is no longer just simply '*not* quantitative research', but has developed an identity (or maybe multiple identities) of its own.

Despite the multiplicity of approaches to qualitative research, some common features of qualitative research can be identified. Qualitative research is intended to approach the world 'out there' (not in specialized research settings such as laboratories) and to understand, describe and sometimes explain social phenomena 'from the inside' in a number of different ways:

- By analyzing experiences of individuals or groups. Experiences can be related to biographical life histories or to (everyday or professional) practices; they may be addressed by analyzing everyday knowledge, accounts and stories.
- By analyzing interactions and communications in the making. This can be based on observing or recording practices of interacting and communicating and analyzing this material.
- By analyzing documents (texts, images, film or music) or similar traces of experiences or interactions.

Common to such approaches is that they seek to unpick how people construct the world around them, what they are doing or what is happening to them in terms that are meaningful and that offer rich insight. Interactions and documents are seen as ways of constituting social processes and artefacts collaboratively (or conflictingly). All of these approaches represent ways of meaning, which can be reconstructed and analyzed with different qualitative methods that allow the researcher to develop (more or less generalizable) models, typologies, theories as ways of describing and explaining social (or psychological) issues.

How do we conduct qualitative research?

Can we identify common ways of doing qualitative research if we take into account that there are different theoretical, epistemological and methodological approaches to qualitative research and that the issues that are studied are very diverse as well? We can at least identify some common features of how qualitative

research is done.

- Qualitative researchers are interested in accessing experiences, interactions and documents in their natural context and in a way that gives room to the particularities of them and the materials in which they are studied.
- Qualitative research refrains from setting up a well-defined concept of what is studied and from formulating hypotheses in the beginning in order to test them. Rather, concepts (or hypotheses, if they are used) are developed and refined in the process of research.
- Qualitative research starts from the idea that methods and theories should be appropriate to what is studied. If the existing methods do not fit with a concrete issue or field, they are adapted or new methods or approaches are developed.
- Researchers themselves are an important part of the research process, either in terms of their own personal presence as researchers, or in terms of their experiences in the field and with the reflexivity they bring to the role – as are members of the field under study.
- Qualitative research takes context and cases seriously for understanding an issue under study. A lot of qualitative research is based on case studies or a series of case studies, and often the case (its history and complexity) is an important context for understanding what is studied.
- A major part of qualitative research is based on texts and writing – from field notes and transcripts to descriptions and interpretations and finally to the presentation of the findings and of the research as a whole. Therefore, issues of transforming complex social situations (or other materials such as images) into texts – issues of transcribing and writing in general – are major concerns of qualitative research.
- If methods are supposed to be adequate to what is under study, approaches to defining and assessing the quality of qualitative research (still) have to be discussed in specific ways that are appropriate for qualitative research and even for specific approaches in qualitative research.

Scope of *The SAGE Qualitative Research Kit*

Designing Qualitative Research (Uwe Flick) gives a brief introduction to qualitative research from the point of view of how to plan and design a concrete study using qualitative research in one way or another. It is intended to outline a framework for the other books in *The SAGE Qualitative Research Kit* by focusing on how-to-do problems and on how to solve such problems in the research process. The book addresses issues of constructing a research design in qualitative research; it outlines stumbling blocks in making a

research project work and discusses practical problems such as resources in qualitative research but also more methodological issues like the quality of qualitative research and also ethics. This framework is filled out in more detail in the other books in the *Kit*.

Three books are devoted to collecting or producing data in qualitative research. They take up the issues briefly outlined in the first book and approach them in a much more detailed and focused way for the specific method. First, *Doing Interviews* (Svend Brinkmann and Steinar Kvale) addresses the theoretical, epistemological, ethical and practical issues of interviewing people about specific issues or their life history. *Doing Ethnography* (Amanda Coffey) focuses on the second major approach to collecting and producing qualitative data. Here again practical issues (like selecting sites, methods of collecting data in ethnography, special problems of analyzing them) are discussed in the context of more general issues (ethics, representations, quality and adequacy of ethnography as an approach). In *Doing Focus Groups* (Rosaline Barbour) the third of the most important qualitative methods of producing data is presented. Here again we find a strong focus on how-to-do issues of sampling, designing and analyzing the data and on how to produce them in focus groups.

Three further volumes are devoted to analyzing specific types of qualitative data. *Using Visual Data in Qualitative Research* (Marcus Banks) extends the focus to the third type of qualitative data (beyond verbal data coming from interviews and focus groups and observational data). The use of visual data has not only become a major trend in social research in general, but confronts researchers with new practical problems in using them and analyzing them and produces new ethical issues. In *Analyzing Qualitative Data* (Graham R. Gibbs), several practical approaches and issues of making sense of any sort of qualitative data are addressed. Special attention is paid to practices of coding, of comparing and of using computer-assisted qualitative data analysis. Here, the focus is on verbal data, like interviews, focus groups or biographies. *Doing Conversation, Discourse and Document Analysis* (Tim Rapley) extends this focus to different types of data, relevant for analyzing discourses. Here, the focus is on existing material (like documents) and on recording everyday conversations and on finding traces of discourses. Practical issues such as generating an archive, transcribing video materials and how to analyze discourses with such types of data are discussed.

Three final volumes go beyond specific forms of data or single methods and take a broader approach. *Doing Grounded Theory* (Uwe Flick) focuses on an integrated research programme in qualitative research. *Doing Triangulation and Mixed Methods* (Uwe Flick) addresses combinations of several approaches in qualitative research or with quantitative methods. *Managing Quality in Qualitative Research* (Uwe Flick) takes up the

issue of quality in qualitative research, which has been briefly addressed in specific contexts in other books in the *Kit*, in a more general way. Here, quality is looked at from the angle of using or reformulating existing criteria, or defining new criteria for qualitative research. This book examines the ongoing debates about what should count as defining 'quality' and validity in qualitative methodologies and examines the many strategies for promoting and managing quality in qualitative research.

Before I go on to outline the focus of this book and its role in the *Kit*, I would like to thank some people at SAGE who were important in making this *Kit* happen. Michael Carmichael suggested this project to me some time ago and was very helpful with his suggestions in the beginning. Patrick Brindle, Katie Metzler and Mila Steele took over and continued this support, as did Victoria Nicholas and John Nightingale in making books out of the manuscripts we provided.

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About this book

Much of current qualitative research is based on approaches using single or stand-alone methods and approaches. Interviews (see Brinkmann and Kvale, 2018) or focus groups (see Barbour, 2018) are often sufficient for doing a study and answering its research questions without using other methods. In ethnography (see Coffey, 2018) a single approach is used, which can include interviewing alongside observation and other methods. Discourse analysis (see Rapley, 2018) and visual research (see Banks, 2018) have their own focuses linked to a specific methodological approach. The general and extensive attention attracted by the idea of using mixed methods shows that the contexts are growing in which a single-method approach is no longer (seen as) sufficient. The concept and use of mixed methods is one focus of this book. However, a look at the history of qualitative research and at practice within qualitative research reveals that combinations of methods are not limited to linking qualitative with quantitative research and that other concepts may take a broader approach. Triangulation is one of these concepts and is the other focus of this book. In the first edition of the *SAGE Qualitative Research Kit*, triangulation was discussed mainly as a way of extending the quality of qualitative research. Therefore, it was one of the topics of the first edition of *Managing Quality in Qualitative Research* (Flick, 2018a). But the trends and methodological discussions mentioned above show that the attention to triangulation and mixed methods now goes beyond enhancing the quality of research and focuses on planning, designing and doing qualitative research (including data collection and analysis). For these reasons, this volume has been added to the books in the second edition of the *SAGE Qualitative Research Kit*. In this sense, the book has two functions in the context of the *Kit*: as a stand-alone book it aims at giving a comprehensive account of the problems and solutions in the field when combining methodological approaches in or with qualitative research; as an addition to the other books in the *Kit*, it rounds off the framework provided by the other books at a methodological level.

Why triangulation and mixed methods in qualitative research?

Chapter objectives

After reading this chapter, you should know:

- the background of triangulation;
- why it was originally introduced in social research;
- what is triangulation and what is not;
- the background of mixed methods;
- why it was originally introduced in social research;
- what is seen as mixed methods and what is not; and
- what is necessary for a first orientation about the issues unfolded in the rest of the book.

Triangulation and mixed methods in qualitative research

Triangulation and **mixed methods** are concepts that refer to a similar idea but make it concrete in differing ways. Both are mainly recognized for combining more than one methodological approach in a study. Mixed methods are focused on combining qualitative and quantitative methods. Triangulation is broader in the kinds of methods that are combined. It is often used as a combination of several qualitative methods. At the same time, it has a broader focus for what is combined – for example, several theoretical approaches or several kinds of data. Triangulation and mixed methods are mostly discussed in alternative and separate discourses, although both discourses can profit from each other. Both concepts can be helpful for extending qualitative **research design**s (see also Flick, 2018b) for several purposes ranging from extra knowledge to confirming research findings.

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Triangulation in the history of qualitative research

Triangulation is a concept that is often taken up in qualitative research when issues of quality are discussed (see also Flick, 2018a). The major link between triangulation and quality of qualitative research is that triangulation means extending the activities of the researcher in the process beyond what is 'normally' done – for example, by using more than one method. However, triangulation is not limited to promoting or assuring the quality of qualitative research, but it is also used to extend the range of insights and knowledge produced in a qualitative study. Both of these aims also apply to mixed methods research, which focuses on combining qualitative and quantitative research and thus extending the research approach and making it more valuable.

The different ways of extending research activities with the aim of extending knowledge will be unfolded in this and the following chapters in some detail. To do this, we will address the theoretical and conceptual basis of triangulation in this chapter before the use of different forms of triangulation is discussed in the following chapters. Different aims, and sometimes myths and reservations are linked to triangulation. Sometimes it is discussed when qualitative research is combined with quantitative approaches in order to give its results more grounding. In general, discussion of triangulation in qualitative research began in the 1970s when Norman Denzin (1970) presented a more systematic conceptualization of triangulation. If we go back a little further in the history of qualitative research, we will find that many of the studies seen as classic studies in qualitative research have not used the concept explicitly but were run according to the principles and practices of what is now discussed as triangulation. We may also find that such practices of triangulation can be seen as a feature of qualitative (also including the use of quantitative) research, as some examples may show.

If we go back to early examples of studying social problems in the history of social research, we find works such as the 'Polish Peasant' or the 'Marienthal' study (see below). Here, we see the combination of several methodological approaches as a starting point for unfolding problems such as unemployment or migration. Combining perspectives on a social problem, William F. Whyte's (1955) classic ethnographic study of a street gang in a major city in the eastern USA in the 1940s offers, on the basis of individual observations, personal notes and other sources, a comprehensive picture of a dynamic local culture. Through the mediation of a key figure Whyte had gained access to a group of young second-generation Italian migrants. As a result of a two-year period of participant observation he was able to obtain information about the motives, values and life-awareness and also about the social organization, friendship-relations and loyalties of this local culture.

The study by Marie Jahoda, Paul Lazarsfeld and Hans Zeisel (1933/1971), *Marienthal: The Sociology of an*

Unemployed Community, is one of the classic studies in qualitative research (see also Fleck, 2004). Here, psychological coping with unemployment in a village was studied in the late 1920s after the main employer of its inhabitants went bankrupt. The result is the elaboration of the leitmotif of a 'tired society' as a condensed characterization of the attitude towards life and the day-to-day practices in the village and of different types of practices in reaction to the unemployment (for example, the 'unbroken', the 'resigned', the 'desperate' and the 'apathetic'). Jahoda (1995, p. 121) has summarized the methodological procedures leading to these insights in combining approaches, such as qualitative and quantitative methods, objective facts and subjective attitudes, observations of current practices and historical material and observations with planned interviews. These principles include linking different methodological approaches (qualitative, quantitative, interviews and observation). At the same time, we find different methodological perspectives (objective facts, subjective attitudes, current and historical issues). In describing the study (1933/1971), the authors list the data they used: cadastre sheets for about 500 families, life histories, sheets for documenting the use of time, protocols, school essays, different statistical data, and historical information about the village and its institutions. Accordingly, Lazarsfeld (1960, p. 14) has made the link between qualitative and quantitative data and strategies a principle, at least for this study. According to Lazarsfeld (1960, p. 15), 'three pairs of data' were used for the analysis: 'natural sources' (statistics of library use) and data that were collected for research purposes (sheets of time use); 'objective indicators' (e.g. health statistics) and subjective statements (interviews); and 'statistics and empathic descriptions of single cases'.

Another early example of triangulation of verbal and visual data is the study by Gregory Bateson and Margaret Mead (1942). A remarkable feature of the study is the empirical approach of producing and analyzing more than 25,000 photographs, masses of filmed material, paintings and sculptures on the one hand and using ethnographic conversations about this material on the other hand. Also, in *The Polish Peasant in Europe and America* by Thomas and Znaniecki (1918–20), different sorts of data were combined: 'undesigned records' as well as an exemplary life history produced by a participant for the study. Finally, Morse (2003, p. 190) sees the work of Goffman (e.g. 1989) as an example of applying triangulation without using the term. These examples show that the use of different sorts of data was characteristic of many classic studies at the beginning of qualitative research.

Barney Glaser and Anselm Strauss and their approach of discovering grounded theories were at the core of the renaissance of qualitative research in the 1960s in the USA and in the 1970s in Europe. Not only the methodological works, but also the studies (e.g. Strauss et al., 1964), are influential and instructive. Again we find different hints for the use of what was later named triangulation. Glaser and Strauss suggest the use of

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different types of data, what they call ‘slices of data’ collected in ‘a multi-faceted investigation’ (1967, p. 65). They also suggest using many different sorts of data, whereas Strauss et al. (1964, p. 36) advocate employing different observers in order to increase the **reliability** of observations that were made independently of each other and comparing them.

These examples may show that the triangulation of data sources, of methods and of researchers has a long tradition in various areas of qualitative research, even if the term was not (yet or always) used. These examples also demonstrate that, in the tradition of these studies, triangulation as an empirical approach to fields and issues was employed as an instrument assessing empirical results as well as a way to gain more insights and knowledge in the research.

Triangulation in the discussion about mixed methods

When the recent discussions about mixed methods started, the existing approach of triangulation was reduced to a specific understanding. Here, the confirmation of results from one approach by those of another approach is regularly seen as its major task (see e.g. Bryman, 2004). Greene et al. (1989) have introduced this limited view in the discussion by referring to two major sources. On the one hand the concept of ‘multiplism’ as outlined by Cook (1985) is suggested for when it is unclear which is the best of several methodologies for a study, to triangulate the most useful or promising ones. On the other hand, referring to Mark and Shotland’s (1987) suggestions of purposes for multi-method designs: (a) triangulation, which seeks convergence of findings; (b) bracketing, which seeks a range of estimates on the correct answer (or triangulation with a confidence interval); and (c) complementarity (Greene et al., 1989, p. 257). With the background of these sources, the field that covers triangulation and mixed methods as ways of engaging multiple (methods) approaches is charted as follows: triangulation is limited to obtaining ‘convergence, corroboration, correspondence of results’ and juxtaposed against complementarity (aiming at elaboration or enhancement), development (aiming at further **sampling**) and expansion (aiming at more breadth and range) by using multiple methods (see Greene et al., 1989, p. 259 for details).

In the following chapters of this book, we will discuss: first whether charting the field of multiple approaches in this way will underestimate the potential of triangulation in several ways; second, how far it leads to some of the current shortcomings in the discussion about mixed methods (see Flick, 2018c); third, whether it gives up the option of using triangulation as a concept to overcome some of the challenges mixed methods research is

currently facing; and fourth, whether it might reduce the potential of using multiple methods within the context of qualitative research. For this purpose, I will start with a short outline of both concepts – triangulation and mixed methods – in this chapter, which then will be unfolded in more detail in the following chapters.

What is triangulation and what is not?

Why triangulate?

Triangulation attracted interest as a methodological strategy for a long time in social research, in **evaluation**, and qualitative research in particular, as an obligation to use combinations of methods, of data, and of researchers for improving the **validity** of a study and to 'withstand critique by colleagues' (Mathison, 1988, p. 13). This attraction was linked for some time to the combination of convergence of results, their validation and legitimation of research by applying triangulation. However, triangulation very soon became the object of a more or less fundamental critique. Put simply, the concept of triangulation means that an issue of research is considered – or in a constructivist formulation is constituted – from (at least) two points. Normally, the consideration from two or more points is materialized by using different methodological approaches (see Chapter 2). The concept of triangulation was imported from land surveying and geodesy, where it is used as an economic method of localizing and fixing positions on the surface of the earth (see Blaikie, 1991, p. 118). The definition used in this context sees triangulation as a method of locating a point from two others a known distance apart (Clark, 1951, p. 145).

In a more metaphorical sense, Campbell and Fiske (1959) and Webb et al. (1966) introduced triangulation into general methodological discussion in the social sciences. At that point, the idea was already that the issue under study is also constituted by the methods used to study it. At that time, a rather negative reading was dominant: that the issue is possibly biased by the methods that are used and that results have to be seen as artifacts. The leading question was whether 'a **hypothesis** can survive the confrontation with a series of complementary methods of testing' (Campbell and Fiske, 1959, p. 82). This led to considerations of how to prevent such a **bias**, and 'unobtrusive' and 'nonreactive measurement' (Webb et al., 1966) were stipulated. One strategy was the combination of different measurements and methods – in a 'multitrait–multimethod matrix' (Campbell and Fiske, 1959). In this context, the metaphor of triangulation is imported 'from navigation and military strategy that use multiple reference points to locate an object's exact position' (Smith, 1975, p.

273) to the social sciences.

For a better understanding of the concept of triangulation, it might be helpful to see what is *not* meant by it. In combining methods, it does not mean that one method is used for collecting data (e.g. an interview) and another one (e.g. coding) is used for analyzing those data. This is obvious and does not need an extra term. Neither does it mean the exploratory use of qualitative methods before the actual study using a standardized method, if the exploratory study is not seen as a genuine and stand-alone part of the project, but is only used for developing a questionnaire and the results of the first step do not become part of the final results of the whole study.

What is mixed methods and what is not?

Why mixed methods?

The reasons given for this kind of combination – in distinction from triangulation –have a broader range. A major intention in developing this approach was to end a rigorous confrontation between qualitative and quantitative research (paradigms) and to promote a more integrative concept for research combining both fields without too much fundamental and epistemological reflection (Greene, 2007). In this sense, Bryman (1992) identifies several ways of integrating quantitative and qualitative research. Triangulation for him means to check, for example, qualitative against quantitative results. Qualitative research and quantitative research can mutually support each other and provide a fuller picture of the issue under study. They focus on structural (quantitative) or process (qualitative) aspects or link micro and macro levels (see Bryman, 1992, pp. 59–61). All in all, this classification represents a broad range of variants, some of which are determined by the idea that qualitative research captures other aspects than quantitative research and that a combination is based on their distinctiveness. Theoretical considerations are not very prominent in these variants, as the focus is more on research pragmatics.

In more recent publications, (e.g. Hesse-Biber, 2010b) this mixed approach has also been adopted from inside qualitative research (see also Chapter 5).

Creswell (2015, p. 2) has defined what mixed methods is and includes as a core assumption the combination

of quantitative data (statistical trends) and qualititative data (stories and personal experiences) to draw interpretations from both for a better understanding of the issue than with one form of data. He also clarifies what he does not see as mixed methods, such as collecting two sorts of qualitative data or adding qualitative data to a quantitative design as it 'involves the collection, analysis, and integration of *both* quantitative and qualitative data' (2015, pp. 2–3).

Triangulation and mixed methods

Triangulation becomes particularly important in two ways in the context of mixed methods discussions – as an instrument assessing empirical results as well as a way to more insights and knowledge in the research – when training in empirical methods increasingly includes qualitative and quantitative research, and keywords like 'mixed methods' (Tashakkori and Teddlie, 2003a) develop a special appeal. Here again, we should examine critically whether they can really meet the expectations of a pragmatic and at the same time theoretically founded combination of research approaches – in producing knowledge and in assessing the quality of research and results. In this context, the question of how far the concept of triangulation has a special importance in the combination of reflection and pragmatism in mixing methods becomes relevant.

In the following chapters, triangulation as a methodological strategy will be inspected and further spelled out against the background of the research traditions and in the context of the current discussions. Therefore, it should not only become clear what triangulation is and how it can be applied, but also what it is not and what problems can arise from using it in qualitative research. In the second part of this book we will have a similar look at mixed methods as the more recent, complementary and sometimes concurrent concept for combining methodological approaches. Here the focus is on what it can contribute to advance qualitative research.

These brief outlines of the two central concepts discussed in this book are mainly meant to give an orientation. Both will be spelled out in more detail in the following chapters – in their features, their use, distinction and complementarity, first triangulation (Chapters 2–4) and then mixed methods (Chapters 5–6) before, in the final chapters, common issues for both will be discussed (Chapters 7–9).

Key points

- Triangulation has a history in qualitative research that started in the context of quality promotion but has developed further towards producing more and better insights.
- To achieve better insights, it seems necessary to spell out the concept of triangulation in more detail.
- The discussion of the concept has led to more differentiation and to a switch from using triangulation as a strategy of validation to one for more reflection and more knowledge.
- It will be even better if we apply this combination of research perspectives more explicitly as a basis for selecting methods to combine in order to promote quality in qualitative research.
- Mixed methods as an approach has been developed to overcome the confrontation between qualitative and quantitative research(ers).
- Mixed methods is therefore limited to combinations of qualitative and quantitative methods.

Further reading

Triangulation

Triangulation and its theoretical basis are spelled out in more detail in the following sources:

Denzin, N.K. (1989) *The Research Act*, 3rd ed. Englewood Cliffs, NJ: Prentice- Hall.

Flick, U. (1992) 'Triangulation revisited: strategy of or alternative to validation of qualitative data?', *Journal for the Theory of Social Behavior*, 22: 175–97.

Flick, U. (2018) 'Triangulation', in N.K. Denzin and Y.S. Lincoln (eds), *The SAGE Handbook of Qualitative Research*, 5th ed. London and Thousand Oaks, CA: Sage, pp. 444–61.

Mixed methods

Mixed methods and their theoretical basis are explained in more detail in the following sources:

Creswell, J.W. (2015) *A Concise Introduction to Mixed Methods Research*. Los Angeles, CA: Sage.

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Greene, J.C. (2007) *Mixed Methods in Social Inquiry*. San Francisco, CA: Jossey-Bass.

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Glossary

ATLAS.ti

Software for supporting the qualitative analysis of text, images and other data in qualitative research.

Between-methods triangulation

Combination of two independent methods in studying one issue.

Bias

Disturbing influence on research and results.

Communicative validation

Assessment of results (or of data) by asking the participants for their consensus.

Comprehensive triangulation

Combination of the different forms of triangulation (investigator, theory, methods and data) in one mode.

Credibility

Criterion for evaluating qualitative research based on prolonged engagement in the field.

Criteria

Instruments for assessing the quality of research, ideally coming with a cut-off point (benchmark), to distinguish good from bad research.

Data triangulation

Combination of different forms of data.

Deviant case

Case not fitting in or supporting a model or other forms of findings.

Episodic interview

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Interview combining question–answer sequences with narratives (of episodes).

Evaluation

Use of research methods for estimating and deciding about the success of an intervention.

Evidence-based practices

Interventions (in medicine, social work, nursing, etc.) that are based on results of research.

Falsification

Testing theories by trying to show that they are not correct.

Generalization

Transfer of research results to situations and populations that were not part of the research situation.

Hybrid methodologies

Methodologies that include elements from more than one method or approach (e.g. in ethnography).

Hypothesis

In standardized research, assumptions to be tested in research. In qualitative research, hypotheses are used in a more metaphorical sense (e.g. as working hypotheses) without being formulated before the research and being tested.

Indication

Decision about when exactly (under which conditions) a specific method (or combination of methods) should be used.

Informed consent

Participants in a study are informed that they are to be studied and are given the chance to say no to the research.

Investigator triangulation

Combination of more than one researcher either in collaboration or independently to promote the quality of the research.

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Likert scale

Questions in a questionnaire with five (sometimes seven) standardized alternatives for answers, which can be ticked by the respondent.

MAXQDA

Software for analyzing qualitative data; earlier versions were called WinMax.

Member checks

Assessment of results (or of data) by asking the participants for their consensus.

Mixed methods/methodologies

An approach combining qualitative and quantitative methods at a rather pragmatic level.

Narrative-episodic knowledge

Knowledge based on memories of situations and their concrete circumstances.

NVivo

Software for analyzing qualitative data; earlier versions were called ‘Nudist’.

Objectivity

The degree to which a research situation (the application of methods and their outcome) is independent of the individual researcher.

Proposal

Research plan developed for applying for funding or in a PhD or master’s programme.

QDA software

Qualitative data analysis software specially developed to support the analysis of texts like interviews and their coding, administration, etc. Examples are ATLAS.ti, MAXQDA and NVivo.

Quality of life

A concept for analyzing the situation of living in the context of an illness or a treatment, mostly measured

using standardized instruments like the SF-36.

Reliability

One of the standard criteria in standardized/quantitative research, measured, for example, by repeating a test and assessing whether the results are the same in both cases.

Research design

A systematic plan for a research project, including who to integrate in the research (sampling), who or what to compare for which dimensions, etc.

Sampling

The selection of cases, persons, materials, etc., to study from a bigger population or variety.

Semantic-conceptual knowledge

Knowledge organized around concepts, their meaning and relations among each other.

SF-36

A standard questionnaire for analyzing quality of life in the context of a health problem or treatment.

Social representations

A concept for describing the knowledge of social groups about scientific findings or other issues.

Subjective theory

Laypeople's knowledge about certain issues can be organized similarly to scientific theories (e.g. subjective theories of health or illness).

Systematic triangulation of perspectives

The combination of different methods including their (differing) theoretical backgrounds in the study of one issue.

Theoretical sampling

The sampling procedure in grounded theory research, where cases, groups or materials are sampled according to their relevance for the theory that is developed and against the background of what is already

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the state of knowledge after collecting and analyzing a certain number of cases.

Theory triangulation

The combination of different theoretical perspectives in the study of one issue.

Transparency

The degree to which a reader of a research study is enabled to understand how the research went on in concrete terms.

Triangulation

The combination of different methods, theories, data and/or researchers in the study of one issue.

Triathlon

An extreme kind of sport combining running a marathon, swimming and cycling very long distances.

Validity

One of the standard criteria in standardized/quantitative research, analyzed for example by looking for confounding influences (internal validity) or for transferability to situations beyond the current research situation (external validity).

Within-method triangulation

The combination of two methodological approaches (e.g. question–answer and narratives) in one method.

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